

Title of Course:	Cloud Computing
Code and Credit Structure:	IT 457, 3-0-2-4
Course Placement:	BTech (ICT&CS, MnC), MTech(ICT), MSc IT, MSc DS –Elective
Course level:	Under Graduate and Graduate
Instructors:	Prof. P.S. Kalyan Sasidhar, Office: 2109, Faculty Block 2, Extn. 560 Email: kalyan_sasidhar@daiict.ac.in

Aims and Objectives:

Cloud computing systems today, both open source or proprietary are built using a common set of core techniques, algorithms, and design philosophies, all of which are centered around distributed systems. This course will teach the fundamental concepts of how and why Cloud systems work, as well as Cloud technologies that manifest these concepts, such as from Amazon AWS, Microsoft Azure, and OpenStack. Students will learn concepts like virtualization, containers, orchestration and performance analysis at scale. The hands on component of the course is provided through multiple labs focussing on providing the necessary practical experience of building and deploying applications on cloud infrastructure. For example, this will include learning, installing and deploying a web application on Kubernetes platform. The associated laboratory component is designed to expose students to the various aspects of cloud computing.

Course outcomes:

At the end of the course, students are expected to:

1. Understand explain virtualization and its role in cloud/elastic computing.
2. Characterize the distinctions between Infrastructure, Platform and Software as a Service (IaaS, PaaS, SaaS) abstractions, and Public and Private Clouds, and analyze their advantages and disadvantages
3. Program basic distributed algorithms to understand their significance in cloud computing
4. Able to create and deploy simple applications on the cloud

Program outcomes

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X			X	X				X			

P1.Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

P4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

P5.Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

P9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Evaluation Scheme

In-Sem	25%
End sem	25%
Project	30%
Lab	15%
Quizzes	5%

Textbook:

Comer, Douglas. *The Cloud Computing Book: The Future of Computing Explained*. Chapman and Hall/CRC, 2021.

References:

- Buyya, Rajkumar, Christian Vecchiola, and S. Thamarai Selvi. *Mastering cloud computing: foundations and applications programming*. Newnes, 2013.
- Marinescu, Dan C. *Cloud computing: theory and practice*. Morgan Kaufmann, 2022.
- Kai Hwang, Jack Dongarra and Geoffrey Fox, “Distributed and Cloud Computing: From Parallel Processing to the Internet of Things” , Morgan Kaufmann, 2011

Course Conduct and Grading Policy

1. Students are supposed to be present for all lectures and lab sessions.
2. If the attendance falls below 70%, a student will receive an F grade. (An institute-wide detailed policy will be communicated separately and will supersede this, once that takes effect)
3. **All lab assignments must be completed**, failing which zero marks will be awarded for the lab component. One makeup lab session will be scheduled at the end of the semester for students who miss a session due to any reason. For special medical cases, if the Dean, AP has approved their application, additional makeup sessions may be provided to the concerned students.
4. If a case of plagiarism/copying is detected in the exams or lab submissions, then that student will **receive zero marks** in the **complete lab segment (30%)** or the respective **exam component**. The student will also be reported to the Dean AP for possible further actions.

Lecture Outline:

Topics	Number of lectures
I Introduction to Cloud Computing: Introduction, Historical developments, Building Cloud Computing Environments, Operating systems and networks fundamentals	6
II Cloud Computing Architecture: Introduction, Fundamental concepts and models, Cloud Characteristics, Cloud Delivery models, Cloud Deployment models, Economics of the cloud.	6

III Virtualization: Introduction, Characteristics of virtualized environments, Taxonomy of virtualization techniques, Hypervisor, Containers, Docker, Pros and cons of virtualization, Technology examples. Cloud Architecture.	8
IV Principles of Distributed Systems: Elements of distributed systems, Technologies for distributed Computing, consistency and replication.	4
V Service Models in Cloud Computing: Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS) and Metal as a Service (MaaS).	6
VI Real World Cloud Computing Technologies: Introduction to OpenStack, Kubernetes AWS, and Microsoft Azure.	6
VII Guest lectures and demos from Industry experts	5-6

Lab Outline:

Lab 1: Fundamentals of Networking	Socket programming
Lab 2: Introduction to AWS:	The first lab in the series walks the students through the installation and configuration of Virtual Machines on AWS Manager
Lab 2: VPC Configuration	This lab introduces the concept of virtual private clouds and their installation on AWS
Lab3: AWS S3	This lab introduces the concept of Amazon S3. Students will gain knowledge on how to Store and manage data by creating S3 buckets, Protecting data using features like versioning, Hosting a static website directly from an S3 bucket, and Automate storage management
Lab 4: EC2 Instances	This lab introduces how to create EC2 instances and mount an S3 bucket on it to experience how cloud storage and compute can work together seamlessly.
Lab 5: AWS SageMaker	In this lab, students are introduced to a tool for training and deploying ML models.
Lab 6: Kubernetes	This lab introduces containers and automatic orchestration principles using Kubernetes
Lab 7:	This lab introduces the concepts of serverless computing and implement a simple AWS Lambda function using the AWS Management Console. The lab also introduces AWS CloudWatch for observing and debugging serverless applications.